



Automated Game Genre Identification Using Text And Image Classification

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Chapter 1

Introduction

In the past, game genre classification was done through manual effort, which produced inconsistent results as human judgment has differing opinions. However, as time progressed, automated genre detection took the world by storm. It takes both textual and visual classification into consideration. By the use of machine learning algorithms, models analyze the game summaries, thumbnails, screenshots, and icon images to label different genres [3]. What polishes this approach further is the concept of automated tagging, which provides personalized recommendations to the gamers catered according to their tastes [1]. In short, with the evolution of the ever-developing gaming industry, there is a need for automated and dynamic genre classification [4].

1.1 Problem Domain

The final year project falls under the domain Computer Vision, intersecting with natural language processing and machine learning. It addresses the limitations of the previous studies, which either focused on text-based or image-based classification, often restricted to a limited set of genres [3]. By integrating both image and text classification, this project aims to improve genre identification across 23 diverse game types.

To achieve this, the project must overcome the current predicament in previously found imbalanced and fragmented datasets, resulting in incomplete and biased outputs [4]. To address this issue, this project will incorporate MSR's PlayMyDataset, a comprehensive, labeled dataset with vast number of entries that include game summaries, names, id's etc from 4 major platforms (XBOX: 68,796, Nintendo: 22,209, PlayStation: 24,040, PC: 68,796) and gameplay images from three major categories (RTS: 7,166, STSTB: 8,868, Music: 3,916). Hence, genres are well-distributed, allowing the model to overcome common classification challenges like overfitting and underrepresenting common genres.

Building on this foundation, the project's innovation lies in advanced deep learning tech-

techniques like transformers and multilabel pipelines for both visual and textual analysis. By automating the classification process, the system aims to deliver a robust classification model that will revolutionize the gaming industry.

1.2 Research Problem Statement

The scope of game genres has been considerably limited, making it impossible to address the diverse nature of existing video games. Until now, most research focused on only 21 genres [1]. On top of that, synthetic and manual approaches have been used for genre classification in the past [5], which are labor-intensive on a vast dataset and have variable accuracy due to subjective judgments and keyword classifications. Consequently, the narrow focus and limited methods have hindered genre classification in the gaming industry.

Extending on this further, a major challenge is the absence of a publicly available dataset that comprises both visual and textual descriptions. For developing an accurate genre identification and classification system, we need a comprehensive labeled dataset that considers game summaries, screenshots, or gameplay images [3]. The dilemma is the class imbalance issue, which causes overfitting as its models perform well on popular genres, due to which less frequent genres are overlooked [4].

In light of these limitations, Transformers will serve as a good alternative. They have demonstrated exceptional performance in tasks involving natural language processing and visual analysis, but their application to the domain of game genre identification remains largely unexplored. In contrast, traditional deep learning or machine learning models often struggle with understanding context over long text sequences, and they often lead to generalizations on new, unseen data, e.g., SVM or neural network (NN) [5]. Also, a large amount of labeled data is required for training these models, which is resource-intensive to obtain.[1]

1.3 Software Requirements Specification

1.3.0.1 Functional Requirements:

The system will provide a user-friendly web portal with proper orientation so that users can easily navigate through webpages .

- The system shall integrate both gameplay images and textual summaries to classify the 23 genres.

- The system will incorporate deep learning algorithms and transformers-based approaches for developing multilabel pipelines across 23 genres.
- The system shall include an input field to input game summaries.
- The system will include classification scores and criteria for authentic result evaluation.

1.3.0.2 Non-Functional Requirements:

- The system shall include prompt response time for output generation.
- The system shall be interactive and user-friendly.
- The system shall protect user-sensitive information (passwords, email etc).

Chapter 2

Literature Review

For ages, game genre classification has been a hot topic for many researchers, focusing mainly on text-based or image-based classification. Machine learning (ML) and natural language processing (NLP) techniques are considered to be best suited for the classification of game descriptions, reviews, and other text metadata. Other models used in the past relied heavily on traditional manual approaches, such as TD-IDF, due to which they achieved little success by keyword analysis. These methods have lacked in capturing the hybrid genres in a broader context.

Recently, deep learning (DL) algorithms have flourished in text-based classification. Long Short-Term Memory (LSTM) and transformers like BERT, RoBERTA, and TinyBERT capture the semantics between different words. For example, 'real-time strategy in com- bat' is classified based on its whole context through contextual embeddings rather than a single keyword.

Despite these obvious achievements, relying only on text may overlook the important visual elements of video games. Keeping this in mind, computer vision has emerged as a vital tool for classification. Convolutional Neural Networks (CNNs) are used on gameplay screenshots, box art, and icons [3]. Clues like art style and character designs are used for classification [3]. This approach has only fallen short when dealing with similar visual clues, which makes it difficult to classify it without additional context.

The exploration of multimodal learning was a great contribution as it was an innovative approach towards genre classification as it took both text and image into consideration. For example, a shooting game will look into visual objects (e.g., guns, bullets, etc.) or game descriptions. By feature fusion, you combine elements from images and text that capture complementary information, which isn't evident using single fusion [1].

This multimodal classification has been explored extensively [1] until recently, and the area of multilabel classification of game genres is untouched. Most studies have focused on single-label classification, where games are assigned one genre, overlooking the fact

games can span multiple genres.

This project leverages game summaries and gameplay image data, incorporating advanced models like transformers, Large Language Models (LLMs), and CNNs to build a comprehensive multilabel system classifying a wide range of genres. MSR's PlayMyDataset, is a comprehensive and labelled dataset which is going to counter the challenges of overlapping genres. Moreover, the introduction of a multilabel classification system, model will perform better with hybrid nature.

2.1 Related Research

The very recent studies have explored multiclass classification, using a dataset of 50,000 video games [1]. Followed by Character networks using eigenvectors and Laplacian matrix to classify 3 genres (romance, fiction and fantasy). The only issue was the data imbalance factor and complex character interactions [2]. CNN's are used for object detection of 205, 554 images [3]. Furthermore, support vector machines (SVM) were used to categorized the genres based on EEG signals from different gamers [5].

2.1.1 Research Item 1

The foremost research item is "Video Game Genre Classification Based on Deep Learning," by Yuhang Jiang. This paper was published in TopSCHOLAR, which is the institutional repository of Western Kentucky University (WKU) [1].

2.1.1.1 Summary of the research item

The mentioned study proposes an innovative approach to classify video games using deep learning techniques. The study uses an all-encompassing dataset of 50,000 video games that includes cover images, game descriptions, and genres that is sourced from IGDB.com. Three classifications were considered: CNNs for analyzing visual clues, LSTMs, and Universal Sentence Encoders for text-based models, and finally a multi-modal framework that fuses both for further refinement and accuracy.

2.1.1.2 Critical analysis of the research item

Stated Scope : The current dilemma is the absence of a comprehensive classification system, which this paper discussed. It highlights the fact 'why there is a need for automated systems' and how automated systems have expanded. By utilizing

transformer-based approaches, this study has enhanced the classification process by retrieving games with proper organization, making it particularly an area of interest for researchers.

Challenges: This study has mainly faced genre-overlapping challenges, which hinder the accurate classification process. Additionally, the dataset needs to be diverse for training the model so the style and format of video games are captured.

Methodology: Three classification methods are explored: CNNs for capturing the visual clues and visual objects, LSTMs and Universal Sentence Encoders for textual analysis, and finally a multimodal framework that will integrate both modalities. Hence, this approach shows how the same problem is dealt with individually and in combination for better results, and it also shows the effectiveness of deep learning techniques.

Innovation: This paper is novel and innovative in its approach as it leverages the current deep learning algorithms for the discussed domain of game genre identification. Not many people have given attention to it specifically in academia. The dataset is all-encompassing, and the multimodal fusion modal is a clear indication of an achievement in this field, and it also lays the foundation for future work.

Evaluation The research study is concluded with a comprehensive model evaluation using clear performance metrics. Top 1 and Top 3 accuracy rates show the accuracy rates, illustrating how each method has proven to be robust or has fallen flat. The final deductions have proven the multimodal approach has received the highest accuracy.

Limitations and Future Work: This study mentioned some of the obstacles, such as biased datasets and no significant work conducted on multilabel classification. Future work may delve deeper into this domain to explore multi-label classification and to investigate the impact of additional modalities, such as audio features, paving the way for further improvement.

2.1.1.3 Relationship to the proposed research work

The methodologies and findings from the paper "Video Game Genre Classification Based on Deep Learning" directly relate to the approach of automated game genre identification using text and visual classification. By carrying forward the multi-modal approach, which integrates both image-based (CNN) and text-based classification on LSTMs and Universal Sentence Encoder models, we can enhance the accuracy and reliability of our classification system. The challenges of genre ambiguity and multilabel classification highlighted in the paper align with the focus on identifying multiple genres for each game. Furthermore, this paper provides valuable insights of improvement for the approach that is to be offered in the future, particularly in handling genre overlap and bias in datasets.

2.1.2 Research Item 2

The first research item is "Genre Classification using Character Networks," authored by Rahul, Divya Agarwal, Ayush, and Devika Vijay. This paper was published in 2021 at the 5th International Conference on Intelligent Computing and Control Systems held in Madurai, India [2].

2.1.2.1 Summary of the research item

This research paper discusses genre classification in literature, emphasizing the subjective nature of genres and the overlap between them. It highlights the complexity of character interactions within narratives and proposes the use of character networks as a novel approach to improve genre classification accuracy. The author details the process of character network extraction, which involves NLP techniques to identify characters and their interactions, represented as graphs. They explore various features, including stylometric and content-based attributes, to enhance classification performance using machine learning algorithms. The study also references previous works that made use of character networks for emotional polarity. The central premise of the research is to enhance the understanding of character relationships and the structure of narratives with the use of modern neural network models applied to narrative graph structure. Character networks are inferred to be able to capture intricate interactions, and this may provide scope for further studies within the domain of character networks and literary analysis.

2.1.2.2 Critical analysis of the research item

Novel Approach : The use of character networks represents a significant advancement in genre classification. By focusing on character interactions, the authors provide a fresh perspective that moves beyond traditional statistical methods, which often overlook narrative structure.

Comprehensive Methodology : The paper presents a comprehensive methodology on character network extraction which also includes the use of certain Natural language processing (NLP) techniques such as tokenization, lemmatization as well as part of speech tagging . This thorough approach enhances the reliability of the data used for analysis.

Integration of Machine Learning : The research effectively combines character network analysis with machine learning algorithms, demonstrating how these networks can improve classification accuracy. The use of eigenvectors of Laplacian matrices to compare character graphs is particularly noteworthy.

Limited Dataset : The paper does not specify the size or diversity of the dataset used for

testing the proposed method. A more extensive and varied dataset could enhance the

generalizability of the findings.

Complexity of Implementation : While the methodology is robust, the complexity of implementing character network extraction and analysis may pose challenges for practitioners. The paper could benefit from providing clearer guidelines or tools for replication.

Lack of Comparative Analysis : The study would benefit from a comparative perspective with available techniques of genre classification. This would place the efficacy of character networks in perspective with regards to conventional ones.

All in all, the research work makes an effective argument in favor of the character networks in genre classification. Its original approach and combining the machine learning techniques make the work significant in literary studies field. Nevertheless, it would be possible to increase the strength and utility of the results by considering these limitations concerning dataset diversity, complexity of implementation and the nature of genres subjectivity. This, for the sake of completeness, could be the preparation of the stage for a more profound investigation into these subjects and, therefore, more efficient classification systems in a given domain.

2.1.2.3 Relationship to the proposed research work

The research on "Genre Classification using Character Networks" closely relates to our proposed work on Automated Game Genre Identification Using Text and visual classification by offering a novel perspective on genre classification through the analysis of character interactions. While the paper focuses on literature, its exploration of narrative structures and the use of modern machine learning techniques, such as graph-based character networks, provides valuable insights that can be adapted for analyzing game narratives. The challenges faced in dataset limitations and the complexity of implementing these methods highlight areas we can address, particularly by focusing on the multimodal aspects of text and visual data in games and by working with PlayMyDataset which deals with a considerably large amount of data across 4 platforms with many entries (XBOX: 68,796, Nintendo: 22,209, PlayStation: 24,040, PC: 68,796).

2.1.3 Research Item 3

Another research item is "Game Genre Classification from Icon and Screenshot Images Using Convolutional Neural Networks " is authored by Chayanin Suatap and Kar Pataukhom. This paper was published by Chiang Mai University, specifically under the Graduate School and Department of Computer Engineering, Faculty of Engineering.

2.1.3.1 Summary of the research item

This paper investigates the relationship between game genres and their visual representations, like game coins and screenshots. The dataset, which consists of 25,001 Android games across 17 genres, was collected using a web crawler. A CNN architecture was developed and fine-tuned its parameters for optimal performance. An ensemble method was employed, which improved classification accuracy to compare single CNN models. Results for single CNN models achieved top-1 accuracy of 40.3% for icons and 46.7% for screenshots, whereas combined models reached 55.3% accuracy and 8.6% improvement for single image classification. The results clearly indicate computers can effectively learn relationships between graphic assets and game genres to assist game developers with asset selection and creating recommendation-based systems built on visual content. The paper concludes with suggestions for future work.

2.1.3.2 Critical analysis of the research item

Novel Approach: This research paper is relevant in its approach as it assists game developers in enhancing the user experience. The incorporation of CNN is a well-established strategy, but the application of it to game genre classification is a novel angle that adds value to the existing literature.

Dataset and Methodology: The dataset comprises an extensive collection of 25,001 Android games. A web crawler was included to collect data on images from different games. However, the data imbalance factor has hindered the results as certain genres are overrepresented.

Model Performance: For single images, the reported accuracies are 40.3% for icons and 46.7% for screenshots, whereas for combined images, the accuracy is 55.3%. Furthermore, the model outperforms any human testing. Further weight to the result can be added through comparison with state-of-the-art models.

Ensemble Method: The positive aspect of this study is the use of a soft voting ensemble, which is an asset as it combines predictions. However, no elaborations on specific configurations of ensembles and how they were determined are mentioned.

Proposed Future Work: The authors suggest several applications for their model, such as developers who can assist in asset selection and creating recommendation systems catering to users' preferences. The papers present various hints on such perspectives but don't present any case studies to illustrate how such implementations can be put into practice.

Clarity and Structure: This is a well-structured paper, with clear sections outlining the methodology, results, and conclusions. However, some sections could be more concise

and crafted with proper explanations, particularly for learners unfamiliar with deep learning concepts and jargon.

2.1.3.3 Relationship to the proposed research work

The research on "Game Genre Classification from Icon and Screenshot Images Using Convolutional Neural Networks" closely relates to our proposed work on Automated Game Genre Identification Using Text and Image Classification, particularly in its use of CNNs. The integration of screenshots and game icons in research offers valuable insights into visual components for game genre classification, which is a key focus in our current study. Nevertheless, this paper's work is focused on single-label classification, while the proposed work is inclined towards multi-label classification, handling a combination of complex genres across multiple platforms. The data imbalance factors are also highlighted in this paper, due to which some genres are overrepresented. To overcome this, PlayMyDataset is a larger, diverse dataset that puts both text and image into consideration to produce an accurate and balanced genre identification model.

2.1.4 Research Item 4

Another research item is "Deep learning for video game genre classification" is authored by Yuhang Jiang and Lukun Zheng. The paper was published by Springer Science+Business Media, LLC, which is part of Springer Nature.

2.1.4.1 Summary of the research item

This paper, "Deep Learning for Video Game Genre Classification," presents a multimodal framework that integrates image and text data to classify video game genres. The authors evaluate the state-of-the-art image-based models. In addition, they utilize transfer learning with pre-trained models like MobileNet and ResNet for image analysis, while for textual analysis, NLP plays an integral role. This study also highlights the challenges in genre classification, such as the diversity of game designs and genres, and emphasizes the need for more resources to improve classification accuracy. The paper concludes with deductions from a comprehensive analysis of experimental results and lays the groundwork for future research work leaning towards multi-label classification.

2.1.4.2 Critical analysis of the research item

Innovative Approach: The integration of both visual (cover images) and textual analysis (descriptions and titles) data into a multi-modal framework is a notable achievement.

This is a clear-cut tactic that applies the strengths of both modalities, leading to improved classification accuracy.

Wide-Ranging Dataset: The substantial dataset of 50,000 video games is a valuable resource for future research. The availability of diverse data types (images, text, and genre information) enhances the robustness of the study. Even though the dataset is vast, there is a risk of overfitting. An in-depth analysis of model generality and validation is not provided or discussed.

Model Evaluation: This paper provides an all-encompassing evaluation of various state-of-the-art models for both visual and text data. This analysis helps establish a benchmark for future studies. However, the authors do not dig deeper into specifics on the methodologies or technologies that could be explored.

Overall, the paper makes a valuable contribution to the field of video game genre classification through its innovative multi-modal approach and comprehensive evaluation of models. However, addressing the limitations related to model interpretability, overfitting, and more specific future work suggestions could enhance the study's impact and applicability in real-world scenarios.

2.1.4.3 Relationship to the proposed research work

The paper "Deep learning for video game genre classification" coincides with the objective of the proposed research, as both stand for the same focus on improving game genre identification using multimodal approaches. The proposed research on Automated Game Genre Identification Using Text and Image Classification will integrate both gameplay images and game summaries for improved accuracy. In contrast, the proposed project aims to extend beyond by addressing the multi-label classification, which the paper briefly mentions but does not fully explore. Additionally, transformers will further enhance the identification accuracy. Thus, this paper will be used as a foundation to build a state-of-the-art multilabel classification system.

2.1.5 Research Item 5

Lastly the research item “Attention guided EEG based Game Genre Classification” is acquired from google scholar.

2.1.5.1 Summary of the research item

This research item makes use of electroencephalography(EEG) for the classification proces. It mentions how fundamental design concepts affects the gamers’ perception of involvement and excitement. EEG signals may contribute to the variations in brain activity when interacting with various genres. Moreover, this study uses features derived from domain characteristics and sample entropy from EEG signals. Support Vector Machines (SVMs) and Neural Networks (NN) are used, out of these two models NN model performs better with an accuracy of 98.35 % relative to 96.24 % for SVM. Hence, it can be rightly judged that EEG can be utilized for finding gamer’s preferences.

2.1.5.2 Critical analysis of the research item

Innovative Approach: The integration of EEG signals to assess cognitive attention with reference to distinct game genres open a new portal to user players’ point of view and preference. In addition, game developers may leverage from this, such that they’ll develop games tailored to the users preferences.

Comprehensive Methodolgy : By employing machine learning algorithms, such as SVM and NN, the effectiveness can measures with proper and comprehensive results. the stated accuracy rates indicate a strong performance (98.35 % for NN and 96.24 % for SVM).

Feature Selection : This study emphasizes on the concept of frequency domain characteristics and Sample Entropy as features for classification. What would add more weight is combination with other physiological parameters for added accuracy.

Generalizability : One of the shortcomings of this paper is the limited number of genres it has worked on comprising only of Action and Puzzle. This dosent capture the vast and ever evolving gaming spectrum.

To sum up everything that has been stated so far, this study has laid foundation for incorporating EEG in game genre classification. Moreover, the scope needs to be widened and further implementation of feature selection needs to be considered with wide-ranging game genres.

2.1.5.3 Relationship to the proposed research work

The research item “Attention guided EEG based Game Genre Classification” aim is to classify game genres based on cognitive responses, enhancing user involvement. Although, this approach is centered towards brain activity, it stressed upon a multimodal framework. Despite the distinct datasets, both approach share the common goal of improving the classification process to produce accurate results. User’s perception is considered in EEg-based paper while the proposed work will look beyond users perception to automatedly classify games into multiple genres by employing transformers and deep learning models.

2.2 Analysis Summary of research items

Table 1: Summary Analysis of Research Items

Paper name	Strengths	Weaknesses
[1]	• Multimodal Approach for image (MobileNet, ResNet, and Inception) and text (RNN, universal sentence encoder) will ensure validated and extensive outputs. • Comprehensive dataset on 50,000 android games to classify wide ranging 21 genres.	• Worked on vast, but still the games were not classified into multiple genres using multiclass classification. • Nobody utilized transformer-based approaches for textual analysis. • Some genres are not well represented, leading to an uneven distribution. For example, genres like "Arcade" are often misclassified because they overlap with other genres, such as "Shooter" and "Indie."
[2]	• Enhanced reliability NLP techniques such as tokenization, lemmatization, and speech tagging. • Empirical validation • Moves beyond traditional statistical methods, that overlook narrative structure. • Combination of character networks + Machine learning algorithms leads to better explanation of the approach.	• Data imbalance leads to biased models that perform well on the majority classes but poorly on minority classes. • If character network data is sparse, the model's ability to discern meaningful patterns can be compromised. • Character interactions might overlook other important aspects of games that contribute to genre classification, such as gameplay mechanics, storyline, or visual style.
[3]	• CNNs adds value to existing	• The study focuses on an

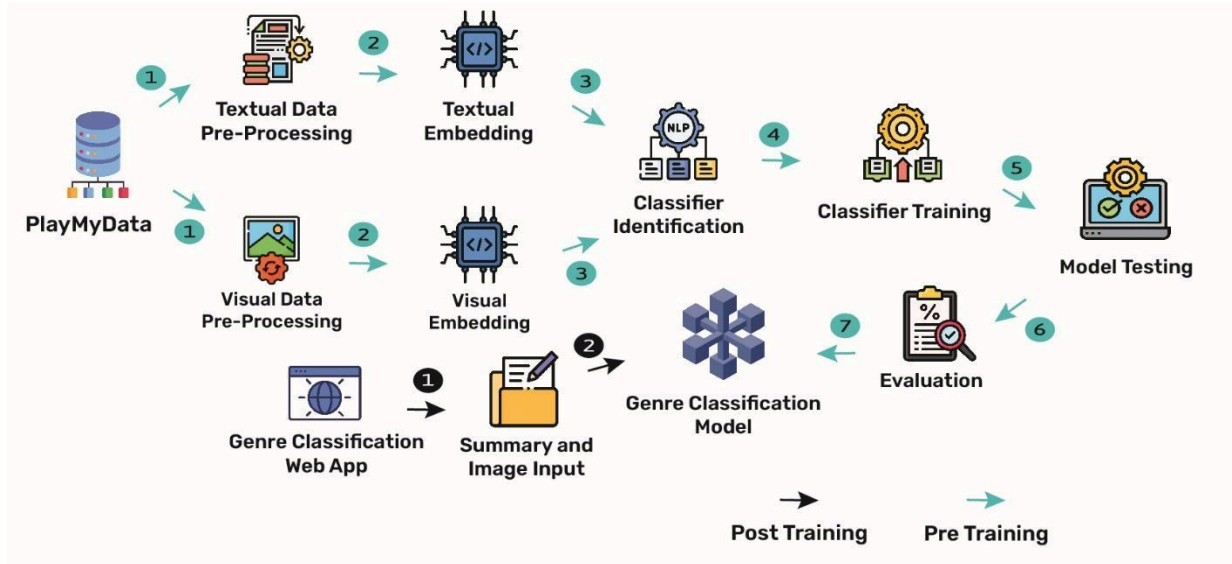
	literature. • Model performance for both single images and combination of images is record through CNNs to find which model performs better. • Use of soft voting ensemble can improve model performance by combining predictions.	Android game dataset, which may limit the generalizability of the findings to other platforms or types of games. • The classification relies heavily on visual features from icons and screenshots, which may not capture the full context or gameplay experience of a game. • The models struggle with specific game genres, such as racing and sports, due to insufficient training data.
[4]	• Multimodal framework captures nuanced information that single-modality models miss. • Diverse dataset of 50,000 video games that allows training and validations of models • Applied proper evaluation metrics like accuracy, precision, f1 score and recall.	• If the dataset is not diverse or comprehensive enough, it can lead to overfitting or poor generalization. • There is a class imbalance issue: If some video game genres are underrepresented in the dataset, the model may perform well on the more frequent genres but poorly on the less common ones.
[5]	• Comparative analysis between NN and SVM for validation and lay ground for future work. • Relevant feature selection.	• Classification of limited set of genres (Action and Puzzle). • The paper likely mentions challenges in optimizing hyperparameters (such as, regularization, and model architecture) for both SVM and neural networks.

Chapter 3 Proposed

Approach

This section emphasizes the approaches adopted to address the limitations/gap in the existing literature. Mainly, NLP and Computer Vision techniques are going to be employed to classify visual and text data for multilabel classification.

Figure 1: Work Flow Diagram



Multimodal Classification Framework

The proposed approach is to develop a multimodal framework that combines both game summaries and gameplay images. The reason behind this combination is to accurately identify games across 23 diverse genres. The core components included are transformers for analysis of game descriptions or summaries and CNNs and YOLO for game-

play images. The original plan is to use the dataset to train the model on candidate pipelines(BERT, RoBERTA, and TinyBERT) for comparative analysis.

Addressing Dataset Challenges

The major challenge faced by past research studies conducted was the imbalanced and fragmented nature of datasets for available game genres. In solution to this problem encountered, the dataset was acquired recently from MSR's proceeding (PlayMyDataset) which is a vast and labelled dataset. It will be used to train the model to eradicate the issue of model underrepresenting the uncommon genres and overfitting.

Multilabel Classification Pipelines

The primary objective of this project is employ a multilabel classification system to classify single game to multiple genres. This will done by taking both modalities into consideration – gameplay images and game summaries – to accurately deduce the genres. For text-based classification the plan is to finetune models like BERT, RoBERTA, and TinyBERT, while for image CNNs and YOLO will be used for comparative analysis.

Model Evaluation/Comparative Analysis

After model is trained and fine-tuned on candidate pipelines, then model will be evaluated based on performance metrics like accuracy, f1 scores, precision and recall. Based

3. Proposed Approach

on the mentioned metrics, the aim is to find optimal combination of achieving robust genre classification. The smart approach is always to fine-tune the models to deal with chances of class imbalance.

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